

Traverse

Report – 2

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Traverse

In the field of surveying, traversing is a technique utilized to establish control points through the measurement and documentation of distances and angles among a sequence of stations. This method is fundamental to numerous engineering and land surveying endeavors, where exact measurements are essential for tasks such as mapping, construction, and boundary delineation. Traversing allows surveyors to create a network of interconnected points, facilitating the precise representation of features across both extensive and limited areas (. Wolf & Ghilani, 2002)

Type of traverse

There are two types of traverse which are listed below:

1. Open Traverse
2. Closed Traverse

Open Traverse

An open traverse refers to a surveying technique in which a sequence of interconnected points is determined by measuring distances and angles, without the necessity of returning to the initial point or linking to another established location. In contrast to a closed traverse, this method lacks a loop or concluding verification, which complicates the identification and correction of measurement inaccuracies. Open traverses are frequently employed in scenarios where it is unfeasible or impractical to complete the traverse, necessitating a heightened degree of precision in measurements to reduce the potential for cumulative errors(Uren & Price, 2018).

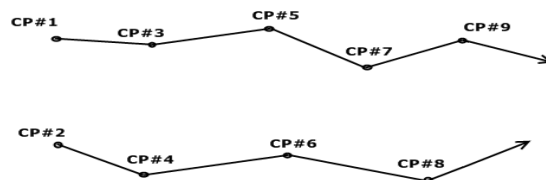


Fig: Open Traverse

Open traverses are extensively utilized in linear engineering endeavors, including roads, railways, pipelines, canals, and river surveys. These projects often span considerable distances along a straight or gently curving trajectory, rendering it impractical or redundant to return to the initial point. In these contexts, open traversing offers an effective approach for determining the position and alignment of the infrastructure, acting as the fundamental control for precise mapping and subsequent construction activities.(Wolf & Ghilani, 2002).

The primary benefits of an open traverse are its efficiency and suitability for elongated projects, facilitating quicker advancement than attempting to close the survey. Nonetheless, this method presents significant drawbacks, including the challenge of identifying or rectifying errors due to the absence of a closure check. Mistakes occurring in the initial phases of the survey can compound and adversely affect the final coordinates. Consequently, meticulous observation, accurate instruments, and frequent verification of measurements are crucial to uphold satisfactory accuracy levels.(Uren & Price, 2018)(Wolf & Ghilani, 2002)

Close Traverse

A closed traverse represents a surveying technique wherein a sequence of measured lines, comprising distances and angles, creates a loop that either returns to the initial starting point or links to another established, known location. This methodology allows surveyors to conduct error assessments via angular and linear closure, thereby confirming that the collected data corresponds with anticipated coordinates. Due to its capacity to validate and correct measurement inaccuracies, closed traverses are frequently favored in high-precision surveying applications(Uren & Price, 2018). Closed traverse surveying is widely used in civil engineering and geomatics. It's commonly applied in setting up construction sites, where control points from a closed traverse help with detailed layout work. They're also important for topographic mapping in small to medium areas that need precise control points

Additionally, closed traverses are crucial for boundary surveys to accurately define land ownership, as legal matters demand high positional accuracy. Surveyors must minimize errors which are achievable through the error-checking features of closed traverses. Another key use is in creating geodetic control networks, which serve as reference points for other surveying tasks in an area. Since closed traverses can check both angular and linear errors, they are a reliable method for developing these precise systems. They're also used in tunnel surveying to ensure accurate coordinate matching between two underground ends, providing consistent data collection across various applications.(Ghilani & Wolf, 2012)



Fig: site for close traverse

Even though the closed traverse method has a lot of benefits, it also has some drawbacks. One of its biggest perks is that it can find and fix measurement mistakes, making it way more precise than open traverses. The closure acts as a math check to ensure the work is accurate, and if there's a tiny error, it can be spread out evenly across the traverse. This gives a lot of trust in the final coordinate values, which is super important for legal and engineering stuffs. Plus, closed traverses can be shaped in different ways, so surveyors can change the layout to fit the land, and they're less impacted by obstacles since control points can be placed smartly(Schofield & Breach, 2007). But these benefits come with a price: they take more time and resources. Finishing a closed traverse usually means going back to where you started or reaching another known control point, which isn't always easy, especially in cities or rough areas. This can make fieldwork a lot more complicated and time-consuming. Another issue is that it relies on being able to see between the

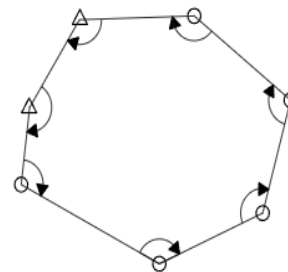


Fig: Closed Traverse

points. In places with thick trees, buildings, or other things in the way, setting up visible stations can be tough and might need more careful planning. Finally, while closed traverses are great for being consistent, their overall accuracy still depends on the starting coordinates. If the starting point isn't exactly right, all the results, even if they seem consistent, could still have some errors (Kavanagh & Bird, 2000).

To sum it up, the closed traverse is a key method in today's surveying world because it's accurate, dependable, and flexible. It's perfect for situations where getting the right position is super important, like setting legal boundaries, planning construction, and building geodetic networks.

The way it allows surveyors to check and fix mistakes through angular and linear closure makes it better than open traverse methods, especially in places where errors can be expensive. Still, surveyors need to think about its downsides, like how much time it takes and the need for good visibility to finish a loop. As technology gets better, with tools like GNSS and total stations, closed traverses are being improved but still play a crucial role in surveying. The mix of accuracy, effort, and use makes the closed traverse a method that, even though it can be challenging, gives great confidence in the results when done carefully. (Uren & Price, 2018)

Advantages of traverse(SP, 2010)

1. High Accuracy (Especially in closed traverse)

In close traverse, the survey starts and ends at a known points, allowing for the detection and correction of measurement errors. This improves the accuracy of both angles and distances and ensures reliable results using errors adjustment techniques like Bowditch rules.

2. Flexibility in the field

Traverse lines can run around the obstacle like buildings, hills or rivers. The surveyor can change direction at each station, making it ideal for obstructed or uneven terrain areas.

3. Effective for long narrow areas

Traverse surveying is especially suited for projects like roads, pipelines, railways, and canals. It allows surveyors to progress steadily along the alignment without needing to maintain the visibility over long distances.

4. Minimal equipment required

The traverse surveying only requires only a theodolite or total station, tripod, tape or EDM. This makes it cost-effective and especially for small to medium-sized projects where advanced technologies may not be required.

5. Internal error checking

In closed traverses, surveyors can calculate and linear misclosure. This means that any small errors occur during the measurement can be detected properly and correctly, ensuring a high level of internal consistency.

6. Reliable for control networks

Traverse surveys are often used to establish control points that serve as the base for detailed topographic or construction surveys.

7. Suitable for legal and construction work

Due to its precision, transverse surveying is preferred for boundary marking, property division, and construction layout in all areas where legal and engineering precision is essential.

Disadvantages of a Traverse(Anderson & Mikhail, 1998)

1. No error check in open traverse

In an open-traverse survey, there is no return to a known starting point, which means that errors cannot be detected, or angle measurements will accumulate and affect the final station.

2. Time Consuming in Closed Traverse

Closed traverse requires returning to the starting point or another control station, which can be logistically difficult and time insensitive, especially in large areas.

3. Skilled personnel required

Accurate angle and distance measurement need a high level of expertise. Mistakes in instrument setup, levelling, or reading can introduce significant errors that affect the entire survey.

4. Complex error adjustments

Although closed traverse allows for error detection (e.g. Bowditch rule) requires additional computation and experience, which can slow down the process.

5. Systematic errors from in accurate start point

If the starting control point has in-accurate co-ordinates or bearing, the entire traverse will inherit this errors, even if the rest of work is measured carefully.

6. Requires clear line of sight

Traverse surveying depends on intervisibility between stations. In urban areas or dense forests, buildings or natural features may obstruct sightlines, making it hard to carry out accurate measurements.

Calculation for northing and easting

calculation for Northing and Easting.

points.	Q.B.	Length	latitude		Depature	
			+	-	+	-
A	88.54°	24.37		9.1	13.26	
B	23.31°	10.85	5.79			10.5
C	68.84°	23.26		26.2		6.31
D	88.46°	27.23		27.007	12.94	
E	14.99°	30.78	42.95		20.24	
F	70.37°	57.04		12.89	54.21	
G	26.57°	41.37		8.93	41.0016	
H	33.96°	29.56	14.59		16.3	
I	37.15°	17.65		33.07		9.21
J	87.95°	38.77		38.76		0.5658
Σ			63.33	79.19	157.95	-26.58

Sum of latitude and Depature:

Sum of latitude = $63.33 + (-79.19)$
 $= -15.86 \text{ m}$

Sum of Depature = $157.95 + (-26.58)$
 $= 131.68 \text{ m}$

for closed traverse

= Sum of latitude = Sum of Depature = 0
 $-15.86 = 131.68 \neq 0$

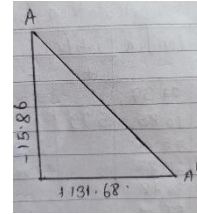
Fig: table showing northing and easting calculation

While checking for the close traverse we can see that the sum of latitudes is not equal to sum of departure is not equal to zero. So, there is an error in the closing.

So, we use Bowditch rules to correct the closing errors.

Calculation of a closing error

- Sum of latitude = -15.86
- Sum departures = 131.68
- Liner error $AA' = \sqrt{15.86^2 + 131.68^2}$
= 132.63m



Perimeter of a traverse = 540.14m

- Closing error = 132.63m in 540.14m
= 1 in 4.07m

Using Bowditch rule to correct the closing error

Correction to latitude for point A = Total latitude correction $\times \frac{\text{Length of a side}}{\text{Perimeter}}$

$$= -15.86 \times \frac{-9.1}{540.14}$$

$$= 0.267\text{m}$$

Correction to a departure for point A = Total departure correction $\times \frac{\text{Length of a side}}{\text{Perimeter}}$

$$= 131.68 \times \frac{13.26}{540.14}$$

$$= 3.23$$

In the same way, all other points for corrected latitude and departure are presented below on the table:

Points	Corrected latitude		Corrected departure	
	+	-	+	-
A	0.267		3.23	
B		0.17		2.55
C	0.76			1.53
D	0.793		3.32	
E		1.26	4.93	
F	0.37		13.21	
G	0.115			0.95
H		0.42	3.93	
I	0.97			2.24
J	1.13			0.13

Corrected latitude and departure for all the points

$$\text{Latitude of point A} = -9.1 + 0.267 = -8.83\text{m}$$

$$\text{Departure of point A} = 13.26 + 3.23 = 16.49\text{m}$$

In the same way all the other latitude and departure is calculated and presented below:

Points	length	Length	latitude		Departure	
			-	+	+	-
A	88.54	24.37	8.83		16.49	
B	23.31	10.85		5.96		13.05
C	68.84	23.26	25.44			7.84
D	88.46	27.23	26.21		16.25	
E	14.99	30.78		44.21	25.17	
F	70.37	57.04	12.52		67.42	
G	26.57	41.37	4.3		40.05	
H	33.96	29.56		15.01	20.23	
I	37.15	17.65	32.1			11.45
J	87.95	38.77	37.63			0.6958
sum			-147.037	65.18	185.61	-33.0358

- Sum of latitudes = $-147.03 + 65.18 = -81.85$
- Sum of departures = $185.61 + (-33.03) = 152.57$

Here, even after the corrected latitude and departures the sum of latitude is not equal to departures.

Here the calculation is not in tolerance

Here are the possible reasons for calculation being out of tolerance which are listed below:

1. Instrumental errors

Total station may not have been properly calibrated, leading to inaccurate angular or distance reading.

2. Human error

Mistakes in reading, recording, or entering measurement can result in significant deviations. Misidentifying stations or incorrect angle setup might be caused during our survey.

3. Environmental factors

Wind and uneven terrain may have affected the line of sight or level stability introducing systematic error in data.

4. Improper application of trigonometry

Error in converting length and bearing to latitude and departure (e.g. using wrong quadrant or incorrect sine/cosine values) leads to incorrect results.

Calculation of a Reduced Level

Step by step process to calculate RL

- The RL of point A = 41.809m (the RL of point A is calculated on week 12 while doing the levelling starting from hive café ending at the station A of a traverse located near library)
- We have a slope distance and a vertical angle of each point.
- Angles are measured from horizontal

$$\text{Elevation difference formula } (\Delta h) = D \cdot \sin(\alpha)$$

Where,

(Δh) = height difference from one point to another point

α = vertical angle in degree

D = slope distance between two point (meter)

Now,

$$RL_{next} = RL_{current} - \Delta h$$

Again,

RL of a point A = 41.809m

Moreover,

To calculate the RL of point B

$$\begin{aligned}\Delta h &= D \cdot \sin(\alpha) \\ &= 1085 \times \sin(1.5) \\ &= 0.217\text{m}\end{aligned}$$

$$\begin{aligned}\text{RL of point B} &= RL_{\text{current}} - \Delta h \\ &= 41.809 - 0.217 \\ &= 41.592\text{m}\end{aligned}$$

In a similar way all the other RL of a point have been calculated and listed below in the table:

points	length	vertical angle in radian	Δh	Reduced Level
A	24.37	1.5	0	41.809
B	10.85	1.5	0.2821	41.5269
C	23.26	1.5	0.60476	40.92214
D	27.23	1.5	0.70798	40.21416
E	30.78	1.5	0.80028	39.41388
F	57.04	1.5	1.48304	37.93084
G	41.37	1.5	1.07562	36.85522
H	29.56	1.5	0.76856	36.08666
I	17.65	1.5	0.4589	35.62776
J	38.77	1.5	1.00802	34.61974

Fig: table showing RL of a traverse point

Personal Portfolio of each member

Hannah Hanid

While working on a closed traverse survey near Roehampton Library, I played a key role in setting up instruments and coordinating measurements with my team. My main tasks involved setting up the total station at three locations and using the reflector prism at two points. I made sure to center the total station carefully with the optical plummet and leveled it accurately with the foot screws. This was important for getting precise measurements of angles.

For the reflector work, I placed the prism over marked spots on the ground and kept it aligned vertically using a bubble level while we measured distances and angles. I communicated closely with my teammates to ensure everything was measured correctly and documented angles, distances, and any adjustments we made.

Even though we faced some challenges like uneven ground and obstacles near the library, we finished the closed traverse with very large errors. This experience really helped me improve my skills with surveying tools, understand how important it is to close angles correctly, and work well in a team. I now feel more confident using what I've learned in class during real fieldwork, and I'm better prepared for future surveying projects.



Chandan Mahara

While conducting a closed traverse survey close to the Roehampton Library, I took on several key tasks that were vital for the accuracy and success of our project. I set up the total station at two locations, ensuring it was perfectly centered over the ground mark with the optical plummet and leveled correctly using the foot screws. I also made sure to orient the instrument properly to start taking measurements from the right reference direction. Getting the setup right was crucial for obtaining accurate horizontal and vertical angle readings throughout the traverse. Additionally, I operated the reflector prism at three different stations, making sure the prism pole was placed directly over the marked points and held straight up, using the circular bubble for precision. I kept in constant communication with the instrument operator to ensure we had a clear line of sight and could take quick, accurate measurements.



Another key task I managed was measuring and recording the height of the instrument (HI) at various stations, which was essential for calculating Reduced Levels (RLs) using vertical angles and slope distance data. Besides my hands-on work, I also served as the data recorder for the team, writing down all angle readings, slope distances, instrument heights, and any necessary corrections or notes in the field book. This role required a lot of attention to detail and organization, as the quality of our field notes directly impacts the accuracy of our calculations after the survey.

Bibek jabegu

During our recent survey project, I was heavily involved in both getting ready for and carrying out a closed traverse. I set up the total station at two important locations, making sure to level and center it properly with a plumb bob and leveling screws. At three other points, I handled the reflector prism, making sure it was right over the marked spots on the ground and kept straight using the bubble level.

A big part of my job was to measure and record the height of the instrument (HI) at each station where we set up the total station. Getting accurate HI readings is super important for figuring out elevations and keeping our data consistent.

I also wrote down all the data we observed, like horizontal and vertical angles, slope distances, and instrument heights, which helped keep our field log neat and organized. Before we started taking measurements, I helped secure each traverse station with wooden pegs, which acted as reference points during the survey.



Rochak Lamsal (Rochu)

I participated in a closed traverse survey near the library in Roehampton, where I handled important technical tasks. My main job was to secure the traverse stations with pegs, which were necessary for creating permanent reference points on the site. I made sure each station was correctly placed and marked, laying the groundwork for all future measurements.

I also set up the total station at three locations, centering it over the pegs with a plumb bob, leveling it with foot screws, and measuring the instrument height. This was crucial for accurate vertical measurements.

At two other locations, I operated the reflector prism, ensuring it was directly above the station point and vertical for precise distance measurement. Additionally, I recorded all field data, including angles, distances, and instrument heights, which were essential for accurate calculations later.



After collecting the data, I helped compute the coordinates for each traverse point using the observed angles and distances, determining the Northing and Easting through traverse techniques. I also calculated the Reduced Level (RL) of each station using vertical angles and slope distances. My role required precision, attention to detail, and a solid understanding of survey principles, providing me with practical experience in using a total station and processing survey data.

CONCLUSION

The survey near Roehampton Library was a practical exercise that improved our understanding of basic surveying techniques. It included tasks like setting up the total station, taking measurements, calculating coordinates, and analyzing errors. This hands-on experience helped us apply what we learned in theory to a real situation, emphasizing important ideas like traverse closure and measurement accuracy. Our group worked carefully to ensure accurate data collection, setting up the total station at three points and using a reflector at two stations for precise measurements.

After gathering the data, we calculated the coordinates and levels of each point, checking for any errors. A crucial part of our analysis was finding the closing error, as a closed traverse should ideally return to the starting point with no error.

Due to limitations in instruments, human mistakes, and environmental factors, we often see small differences. In our case, we noted slight variations in the final northing, easting, and RL values compared to the initial ones. These differences were labeled as closing errors and measured to evaluate the traverse's quality. To fix these errors, we used adjustment methods like the Bowditch Rule, which adjusts the misclosure based on the length of each leg of the traverse. This method assumes that both distance and angle measurements have minor errors that can be corrected.

The adjusted coordinates and RLs gave us a more accurate and consistent view of the traverse, enabling us to move forward with mapping or further analysis if needed. A key benefit of this exercise was enhancing our teamwork and collaboration skills.

Our group of four—Hannah Hanid, Chandan Mahara, Bibek Zabegu, and Rochak Lamsal worked together to share tasks and ensure smooth fieldwork. We switched roles to learn all parts of the process, including setting up the instrument, taking measurements, operating the reflector, and checking data

accuracy. Good communication was essential during the survey, especially when sighting the reflector over long distances and confirming angle readings.

This project emphasized the need for accuracy and careful attention in surveying. Even minor measurement mistakes can add up and greatly affect the final outcomes, particularly in closed traverses where geometric precision is crucial. We learned to thoroughly check instrument settings, ensure the total station is level and centered, and accurately record all measurements. Additionally, we had to consider external factors like changing weather, uneven ground, and obstacles that could impact measurement reliability. Another key lesson was combining field observations with computational analysis. Converting raw measurements into useful spatial data through calculations and error corrections helped us understand how field data contributes to creating precise maps, plans, and construction layouts. This underscores the importance of surveying in civil engineering, urban planning.

In summary, the closed traverse survey project near Roehampton Library was a great learning experience that covered everything from data collection to analysis and reporting. It showed how interconnected surveying tasks are, the importance of accuracy and consistency, and the teamwork needed for successful project completion. This exercise not only strengthened our technical skills but also improved our ability to collaborate, solve problems, and adapt to real-world challenges in the field. These experiences will certainly provide a solid foundation for our future studies and careers in surveying.

Note: Some of the things are taken from my diploma book which I didn't find on the internet so I could not provide a references.

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